Supplementary Notes on Quadratics

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Abstract

This note provides a more detailed, step-by-step derivation of the results presented in Chapter 4 of the lecture notes.

1 Symmetrize Nonsymmetric Matrix

This section answers the question why the following equation holds

$$f(\mathbf{x}) \coloneqq \mathbf{x}^\top \mathbf{A}\mathbf{x} + \mathbf{b}^\top \mathbf{x} + c = \mathbf{x}^\top \tilde{\mathbf{A}}\mathbf{x} + \mathbf{b}^\top \mathbf{x} + c =: \tilde{f}(\mathbf{x}),$$

where $\tilde{\mathbf{A}} \coloneqq \frac{1}{2}\left(\mathbf{A} + \mathbf{A}^\top\right)$ is the symmetric part of $\mathbf{A}$.

To prove this equality, simply substitute the definition of $\tilde{\mathbf{A}}$ into $\tilde{f}(\mathbf{x})$:

$$\begin{aligned}\tilde{f}(\mathbf{x}) &= \frac{1}{2} \cdot \mathbf{x}^\top \left(\mathbf{A} + \mathbf{A}^\top\right)\mathbf{x} + \mathbf{b}^\top \mathbf{x} + c \\ &= \frac{1}{2}\mathbf{x}^\top \mathbf{A}\mathbf{x} + \frac{1}{2}\mathbf{x}^\top \mathbf{A}^\top \mathbf{x} + \mathbf{b}^\top \mathbf{x} + c.\end{aligned}$$

Observe that $(\mathbf{x}^\top \mathbf{A}\mathbf{x})^\top = \mathbf{x}^\top \mathbf{A}\mathbf{x}$. Since $\mathbf{x}^\top \mathbf{A}\mathbf{x} \in \mathbb{R}$, it is a scalar and is therefore equal to its transpose. Hence, $\mathbf{x}^\top \mathbf{A}\mathbf{x} = \mathbf{x}^\top \mathbf{A}^\top \mathbf{x}$ and it follows that $f(\mathbf{x}) = \tilde{f}(\mathbf{x})$. We will use this observation again in future lectures and tutorials.

2 Gradient Derivations

This section explains in details why the following equality holds.

$$\frac{\partial \mathbf{x}^T \mathbf{A}\mathbf{x}}{\partial \mathbf{x}} = (\mathbf{A} + \mathbf{A}^\top)\mathbf{x}. \tag{1}$$

The derivation of the gradient of $\mathbf{b}^\top \mathbf{x}$ is already presented in sufficient detail in the lecture notes, we therefore omit it here.

To show (1), by definition of gradient, we need to find

$$\frac{\partial \mathbf{x}^T \mathbf{A}\mathbf{x}}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial \mathbf{x}^\top \mathbf{A}\mathbf{x}}{\partial x_1} \\ \vdots \\ \frac{\partial \mathbf{x}^\top \mathbf{A}\mathbf{x}}{\partial x_N} \end{bmatrix}.$$

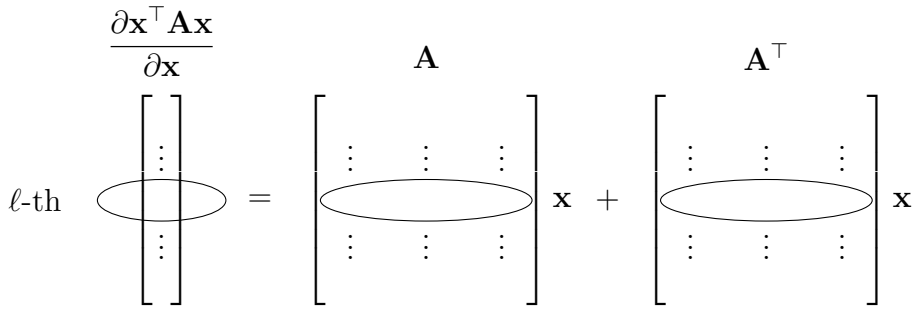


Figure 1: The ℓ-th element of the gradient is the sum of the dot products between $\mathbf{x}$ and the ℓ-th rows of $\mathbf{A}$ and $\mathbf{A}^\top$.

If we can derive an expression for each element of this gradient vector, we then know what the gradient looks like. Consider the ℓ-th element of the gradient vector, where $\ell \in \{1, \ldots, N\}$. By expressing $\mathbf{x}^\top \mathbf{A}\mathbf{x}$ in summation form, we have

$$\frac{\partial \mathbf{x}^\top \mathbf{A}\mathbf{x}}{\partial x_\ell} = \frac{\partial}{\partial x_\ell} \sum_{i=1}^{N} \sum_{j=1}^{N} a_{ij} x_i x_j. \tag{2}$$

Observe that (2) can be decomposed into

$$\frac{\partial}{\partial x_\ell} \sum_{i=j} a_{ij} x_i x_j \quad \text{and} \quad \frac{\partial}{\partial x_\ell} \sum_{i \neq j} a_{ij} x_i x_j.$$

In the first summation, the only term that depends on x_ℓ is $a_{\ell\ell} x_\ell^2$. In the second summation, the only terms that depend on x_ℓ are

$$\begin{cases} x_\ell x_j a_{\ell j}, & \forall j \neq \ell, \\ x_i x_\ell a_{i\ell}, & \forall i \neq \ell. \end{cases}$$

Therefore, by this observation, we have

$$\begin{aligned} \frac{\partial \mathbf{x}^\top \mathbf{A}\mathbf{x}}{\partial x_\ell} &= \frac{\partial}{\partial x_\ell} \sum_{i=j} a_{ij} x_i x_j + \frac{\partial}{\partial x_\ell} \sum_{i \neq j} a_{ij} x_i x_j \\ &= \frac{\partial}{\partial x_\ell} a_{\ell\ell} x_\ell^2 + \frac{\partial}{\partial x_\ell} \left(\sum_{j \neq \ell} x_\ell x_j a_{\ell j} + \sum_{i \neq \ell} x_i x_\ell a_{i\ell} \right) \\ &= 2 a_{\ell\ell} x_\ell + \sum_{j \neq \ell} x_j a_{\ell j} + \sum_{i \neq \ell} x_i a_{i\ell} \\ &= \sum_{j=1}^{N} x_j a_{\ell j} + \sum_{i=1}^{N} x_i a_{i\ell}, \end{aligned}$$

where the last equality is obtained by splitting $2a_{\ell\ell}x_\ell$ into two $a_{\ell\ell}x_\ell$ terms and grouping each with one of the two sumations. This result can be interpreted as the sum of the dot product between the ℓ-th row of $\mathbf{A}$ and $\mathbf{x}$ and the dot product between ℓ-th row of $\mathbf{A}^\top$ and $\mathbf{x}$:

$$\sum_{j=1}^{N} a_{lj}x_j + \sum_{i=1}^{N} x_i a_{i\ell} = \begin{bmatrix} a_{\ell 1} & \cdots & a_{\ell N} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_N \end{bmatrix} + \begin{bmatrix} a_{1\ell} & \cdots & a_{N\ell} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_N \end{bmatrix}.$$

Therefore, we have shown that the ℓ-th elements of the gradients of $\mathbf{x}^\top\mathbf{A}\mathbf{x}$ is equal to the sum of the dot products between $\mathbf{x}$ and the ℓ-th rows of $\mathbf{A}$ and $\mathbf{A}^\top$. See Figure 1 for an illustration. It then follows that (1) holds.